

# Binomial Coefficients in Probability and Expansion

## Answer Key

### Precalculus

#### Quick Answers

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|-------------------------------------|-----------------------------------|
| 1. 15                               | 8. 9                              |
| 2. $x^3 + 6x^2 + 12x + 8$           | 9. (a) the probability = 0.087, — |
| 3. $\frac{5}{16}$                   | 10. (a) the expansion =           |
| 4. 90                               | $1 + 4x + 6x^2 + 4x^3 + x^4$ , —  |
| 5. 56                               | 11. (a) the row sum = 64, —       |
| 6. $16x^4 - 32x^3 + 24x^2 - 8x + 1$ | 12. (a) the coefficient = 6048, — |
| 7. 0.2304                           |                                   |
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#### What is verified

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12 of 16 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 9, 10, 11, 12. The worked solution states what a correct response must contain.

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#### Curriculum

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**Level:** Precalculus

HSA-APR.C.5 – problems 1, 2, 4, 5, 6, 8, 10, 11, 12

HSS-CP.B – problems 3, 7, 9

Difficulty 1–5 of 5 across 16 tagged checks.

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#### Common wrong answers

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1. If they got 30: computed the permutation count and forgot to divide by  $k$  factorial
  4. If they got 10: used the binomial coefficient alone and forgot the power of 3
  7. If they got 0.023: left out the binomial coefficient and counted only one ordering
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1. Evaluate  $\binom{6}{2}$ .

**Solution:**  $\binom{6}{2} = \frac{6!}{2!4!} = \frac{6 \cdot 5}{2 \cdot 1} = \frac{30}{2}.$

15

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2. Expand  $(x + 2)^3$  in descending powers of  $x$ .

**Solution:** Row 3 of Pascal's triangle is 1, 3, 3, 1, and the powers of 2 climb as the powers of  $x$  fall:

$$x^3 + 3x^2(2) + 3x(2^2) + 2^3 = x^3 + 6x^2 + 12x + 8$$

$$x^3 + 6x^2 + 12x + 8$$


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**3.** Of the 32 equally likely sequences of 5 coin flips, 10 have exactly two heads. Find that probability.

**Solution:** Every sequence is equally likely, so  $P = \frac{10}{32} = \frac{5}{16}$ . The 10 is  $\binom{5}{2}$ : it counts which two of the five flips are the heads.

$$P = \frac{5}{16}$$


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**4.** In  $(x + 3)^5$ , find the coefficient of  $x^3$ .

**Solution:** The term with  $x^3$  takes  $x$  from three of the five factors and 3 from the other two, so it is  $\binom{5}{3}x^3 \cdot 3^2 = 10 \cdot 9 \cdot x^3$ .

$$90$$


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**5.** Given  $\binom{7}{2} = 21$  and  $\binom{7}{3} = 35$ , find  $\binom{8}{3}$  using the addition rule.

**Solution:** Each Pascal entry is the sum of the two directly above it:  $\binom{8}{3} = \binom{7}{2} + \binom{7}{3} = 21 + 35$ .

$$56$$


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**6.** Expand  $(2x - 1)^4$  in descending powers of  $x$ .

**Solution:** Take  $a = 2x$ ,  $b = -1$ , coefficients 1, 4, 6, 4, 1:

$$\begin{aligned} (2x)^4 + 4(2x)^3(-1) + 6(2x)^2(-1)^2 + 4(2x)(-1)^3 + (-1)^4 \\ = 16x^4 - 32x^3 + 24x^2 - 8x + 1 \end{aligned}$$

The alternating signs come from the odd powers of  $-1$ .

$$16x^4 - 32x^3 + 24x^2 - 8x + 1$$


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**7.** Five free throws, each made with probability 0.4. Find  $P(\text{exactly } 3)$ , to four decimal places.

**Solution:** One particular ordering of three makes and two misses has probability  $(0.4)^3(0.6)^2 = 0.02304$ , and there are  $\binom{5}{3} = 10$  such orderings:

$$P = 10(0.4)^3(0.6)^2 = 10(0.02304)$$

$$P = 0.2304$$


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8. In  $(x^2 + 3)^3$ , find the coefficient of  $x^4$ .

**Solution:** Each factor contributes either  $x^2$  or 3. To reach  $x^4$  the  $x^2$  must be taken from exactly two of the three factors, leaving one 3:  $\binom{3}{2} (x^2)^2 (3) = 3 \cdot 3 \cdot x^4$ . 9

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9. Five free throws at  $p = 0.4$ . (a)  $P(\text{at least 4})$  to three decimals. (b) Why two terms?

**Solution (a):** “At least 4” means exactly 4 or exactly 5:

$$P = \binom{5}{4} (0.4)^4 (0.6) + (0.4)^5 = 5(0.0256)(0.6) + 0.01024 = 0.0768 + 0.01024$$

$$P \approx 0.087$$

**Solution (b) — instructor-judged.** The binomial formula gives the probability of *exactly*  $k$  successes, one  $k$  at a time. “At least 4” covers two different outcomes, 4 makes and 5 makes, and they cannot both happen, so the two probabilities are disjoint and add. Full credit requires naming the two cases and saying they are disjoint (or mutually exclusive).

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10.  $(1+x)^4$ . (a) Expand in ascending powers. (b) What does the coefficient of  $x^k$  count?

**Solution (a):** Row 4 of Pascal’s triangle is 1, 4, 6, 4, 1, and since one term of the binomial is 1 every power of it is 1.

$$1 + 4x + 6x^2 + 4x^3 + x^4$$

**Solution (b) — instructor-judged.** Multiplying out  $(1+x)(1+x)(1+x)(1+x)$  means choosing, from each of the four factors, either the 1 or the  $x$ . A term  $x^k$  arises once for every way of choosing the  $x$  from exactly  $k$  of the four factors, so its coefficient is  $\binom{4}{k}$  — the number of  $k$ -element subsets of a 4-element set. That is why the coefficients are 1, 4, 6, 4, 1. Full credit requires the choosing-from-factors argument.

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11. Row 6 is 1, 6, 15, 20, 15, 6, 1. (a) Find the row sum. (b) Why is the sum of row  $n$  always  $2^n$ ?

**Solution (a):**  $1 + 6 + 15 + 20 + 15 + 6 + 1 = 64 = 2^6$ .

$$64$$

**Solution (b) — instructor-judged.** Entry  $\binom{n}{k}$  counts the subsets of size  $k$  of an  $n$ -element set, so the whole row counts every subset, sorted by size. A set of  $n$  elements has  $2^n$  subsets, because each element is independently either in or out. Equivalently, substituting  $x = 1$  into  $(1+x)^n$  makes every term equal its coefficient, and the left side becomes  $2^n$ . Full credit for either argument.

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12. **Synthesis.**  $(2x + 3)^7$ . (a) Coefficient of  $x^5$ . (b) Why is the power of 3 equal to 2?

**Solution (a):** The term is  $\binom{7}{5} (2x)^5 (3)^2 = 21 \cdot 32x^5 \cdot 9$ , and  $21 \cdot 32 \cdot 9 = 6048$ . 6048

**Solution (b) — instructor-judged.** Each of the seven factors contributes either its  $2x$  or its 3. Getting  $x^5$  means taking the  $2x$  from exactly five factors, and the remaining two factors must then contribute their 3. The two exponents always add to 7, so an exponent of

5 on  $x$  forces an exponent of 2 on 3. Full credit requires the exponents-sum-to- $n$  reason, not just the correct number.